YouTube Live Streaming Setup

Before you start

You need a YouTube channel with live streaming turned on, plus an encoder (OBS, an ATEM switcher, vMix, or a hardware encoder).

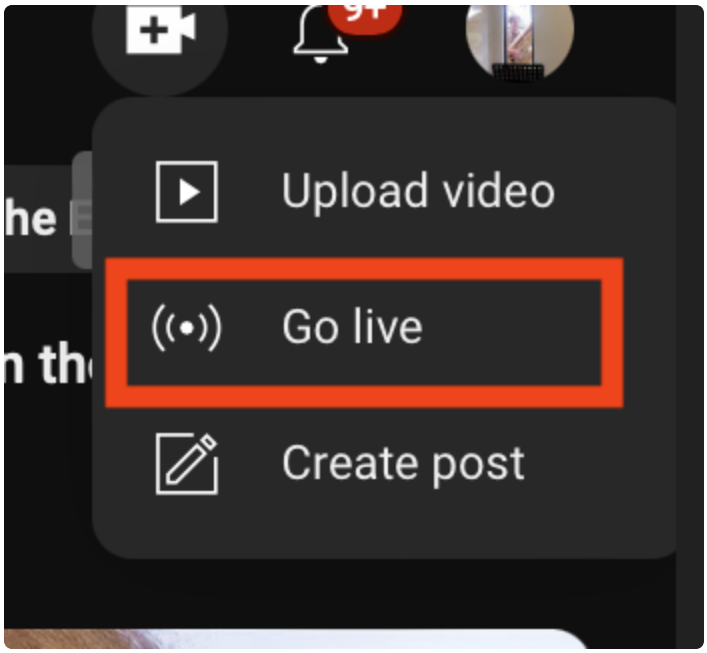
- **Verify the channel.** YouTube asks for phone verification before live streaming is allowed.
- **Enable live streaming early.** The first time you turn it on, YouTube can take up to 24 hours to activate it. Do this at least a day before your first event.
- **Use a computer.** Stream keys and the auto-start/auto-stop toggles are set in YouTube Studio in a desktop browser at studio.youtube.com.

Create the stream in YouTube Studio

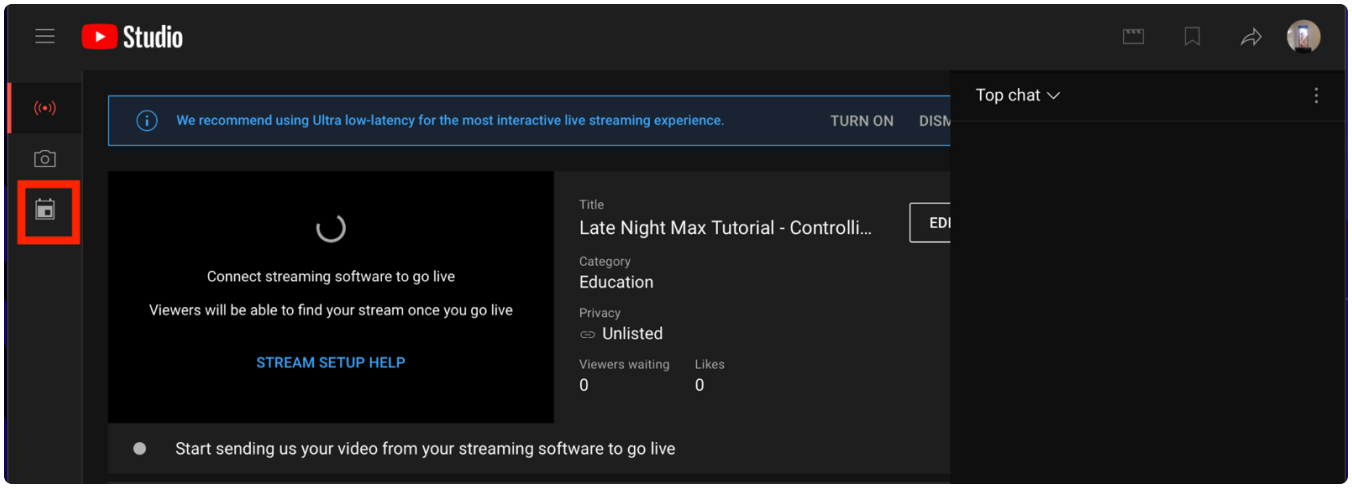
Schedule the stream from the **Manage** tab so auto-start and auto-stop are available and the key stays tied to that event.

Most screenshots below are from an earlier YouTube Studio layout. Buttons may have moved slightly, but the labels are the same.

Step 1. Go to youtube.com, click the camera icon (top right), then **Go live**. This opens YouTube Studio's Live Control Room.



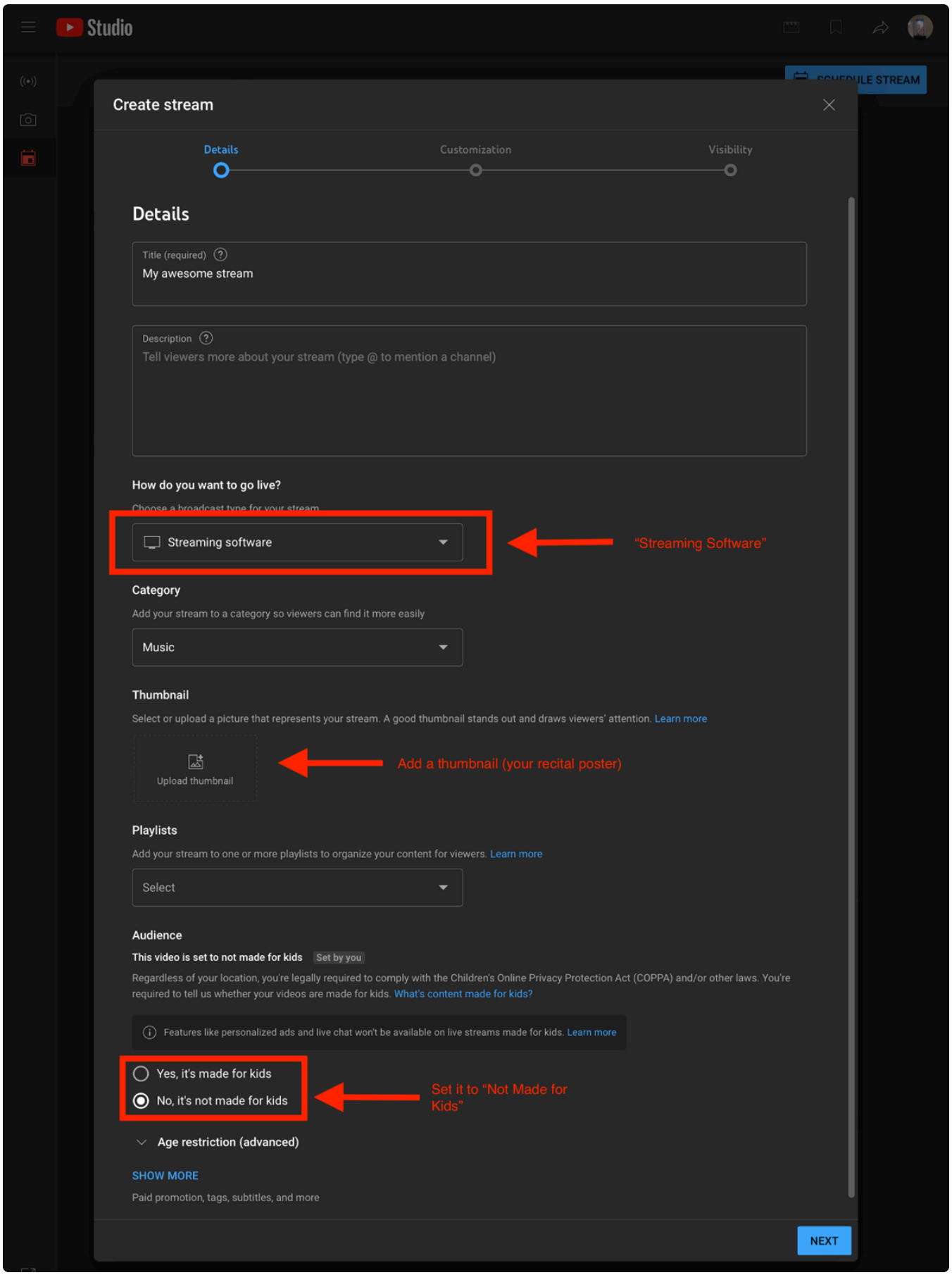
Step 2. In the left sidebar, click **Manage** (the calendar icon).



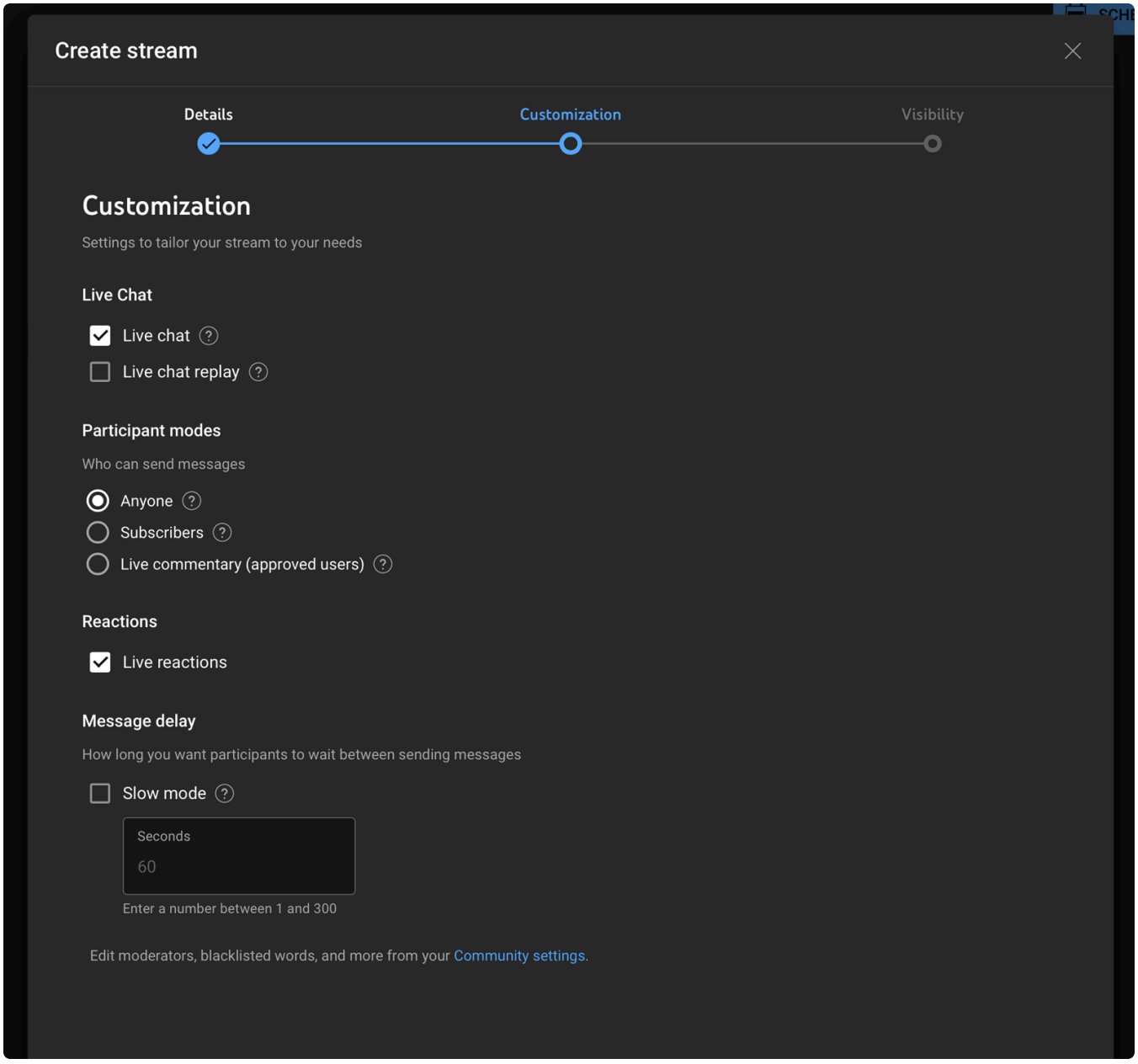
Step 3. Click **Schedule stream** (top right). To copy an earlier stream's title, settings and key, choose **Reuse settings**; otherwise choose **Create new**.



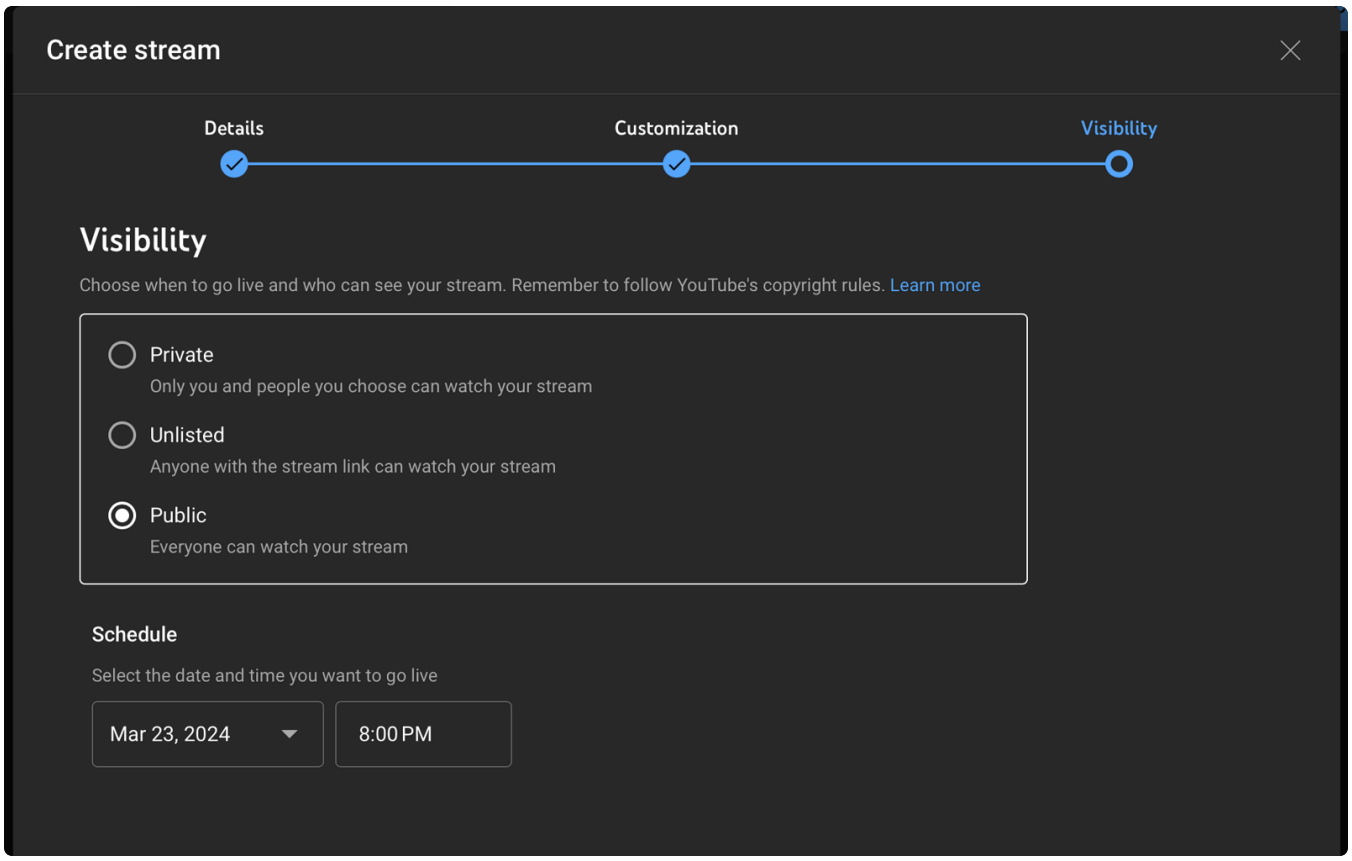
Step 4. On **Details**: add a title, description and category; set *How do you want to go live?* to **Streaming software**; upload a thumbnail; choose **No, it's not made for kids**. Click **Next**.


Studio
Create stream
Details
Customization
Visibility
Details
Title (required)
My awesome stream
Description
Tell viewers more about your stream (type @ to mention a channel)
How do you want to go live?
Streaming software
"Streaming Software"
Category
Add your stream to a category so viewers can find it more easily
Music
Thumbnail
Select or upload a picture that represents your stream. A good thumbnail stands out and draws viewers' attention. Learn more
Upload thumbnail
Add a thumbnail (your recital poster)
Playlists
Add your stream to one or more playlists to organize your content for viewers. Learn more
Select
Audience
This video is set to not made for kids
Set by you
Regardless of your location, you're legally required to comply with the Children's Online Privacy Protection Act (COPPA) and/or other laws. You're required to tell us whether your videos are made for kids. What's content made for kids?
Features like personalized ads and live chat won't be available on live streams made for kids. Learn more
Yes, it's made for kids
No, it's not made for kids
Set it to "Not Made for Kids"
Age restriction (advanced)
SHOW MORE
Paid promotion, tags, subtitles, and more
NEXT

Step 5. On **Customization**, set live chat options however you like. Click **Next**.



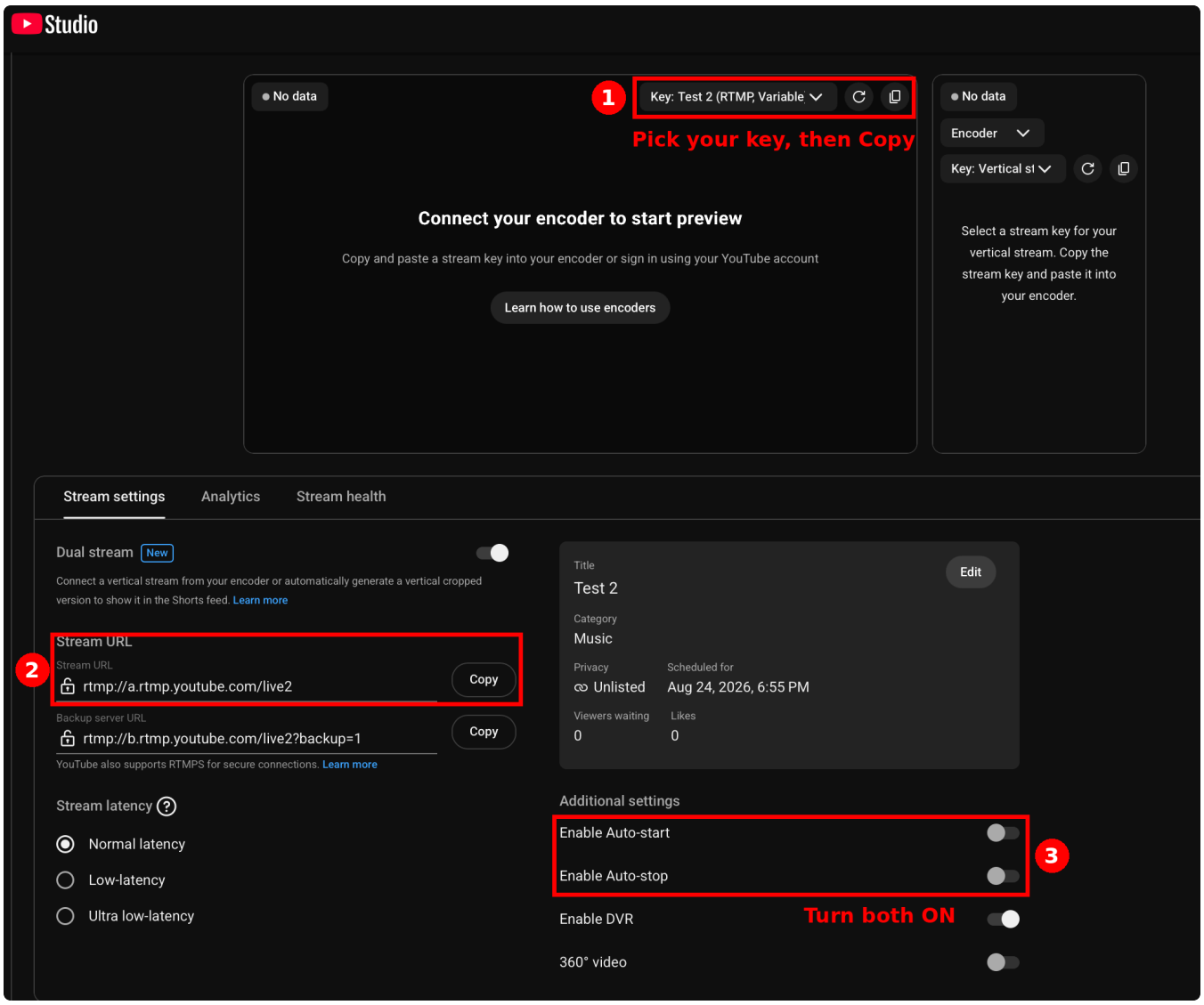
Step 6. On **Visibility**, choose **Public** (or Unlisted/Private), set the date and time of your event, then click **Done**. The stream opens in the Control Room.



Turn on Auto-start and Auto-stop

With both on, your encoder controls the broadcast: start streaming and YouTube goes live; stop streaming and YouTube ends it. No one needs to click **Go live** in the browser.

1. In the Control Room, with your scheduled stream open, find the **Stream settings** tab below the preview.
2. Scroll to **Additional settings**.
3. Turn on **Enable auto-start**. The broadcast goes public as soon as YouTube receives video from your encoder.
4. Turn on **Enable auto-stop**. The broadcast ends and archives as soon as your encoder stops sending.



Box 3: both switches are off by default. Click each one so it turns on (knob slides right).

Setting	What it does	Watch out for
Auto-start	Goes live the moment video arrives	Any test signal you send goes out live, so preview before you hit Stream
Auto-stop	Ends the broadcast when video stops	A network drop or encoder restart can end the stream early; leave it off if you plan to stop manually on an unreliable connection

These toggles carry over when you schedule the next stream with **Reuse settings**.

Copy the stream key

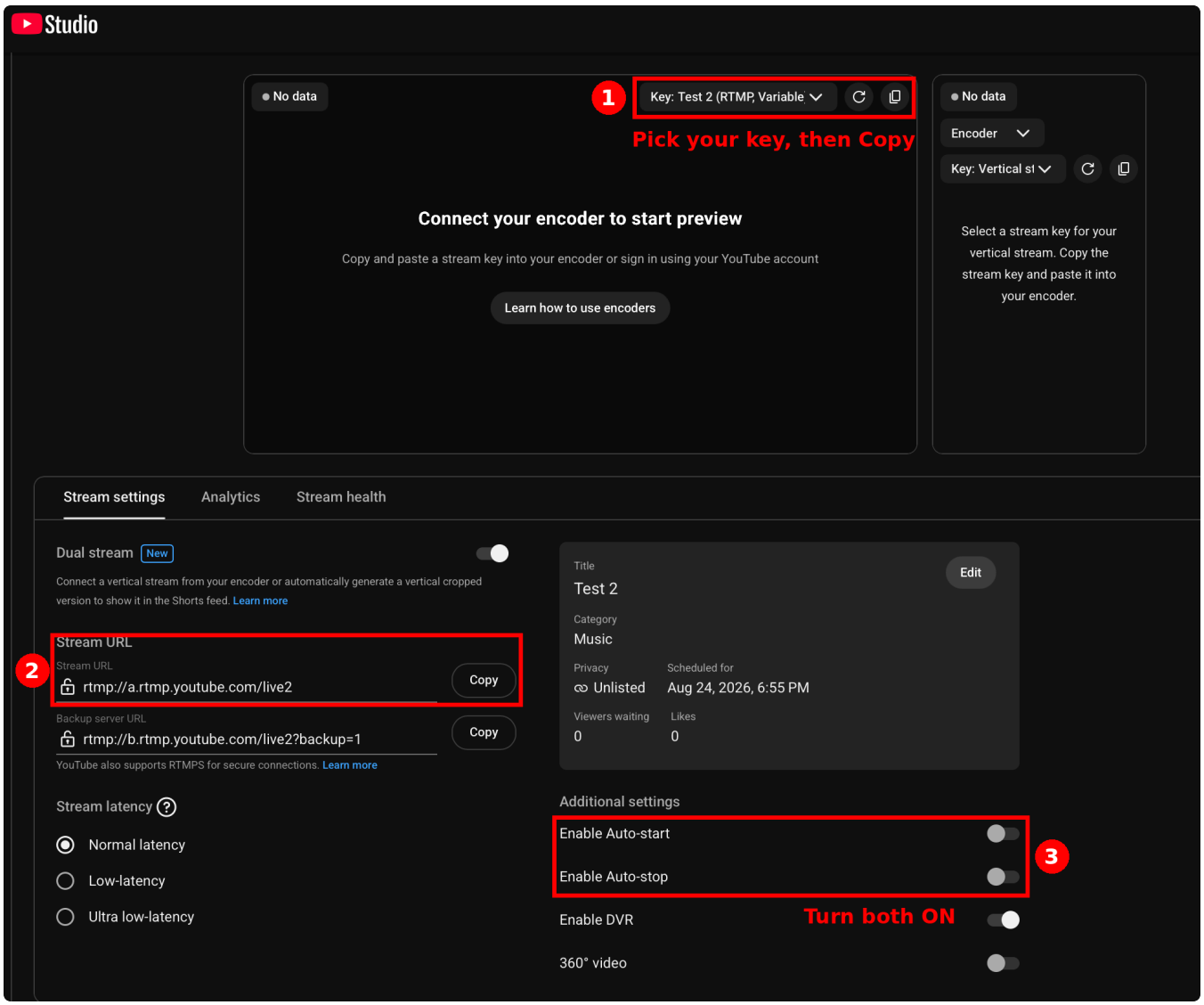
The stream key is what tells YouTube which broadcast your encoder's video belongs to. Use a saved custom key so you only enter it in your encoder once.

Pick or create the key

1. In the Control Room, open your scheduled stream.
2. Open the **Key:** dropdown at the top right of the preview (it shows "Default stream key (RTMP…)" until you pick another).
3. First time only: choose **Create new stream key**, give it a name (for example "Main encoder"), keep **RTMP** and **Variable** bitrate, and click **Create**.
4. Select that key from the dropdown.

Copy it

1. **Box 1:** with your key selected in the **Key:** dropdown, click the **copy icon** (two overlapping squares) right beside it. The full key is now on your clipboard, even though it never appears on screen.
2. Don't click the circular-arrow icon between them: that **resets** the key, and every encoder using the old one stops working.
3. **Box 2:** in the **Stream settings** tab, click **Copy** next to **Stream URL** (`rtmp://a.rtmp.youtube.com/live2`). You don't need to send this one; it's the same for everyone.



If **Dual stream** is on, a second panel on the right has its own vertical-stream key. Leave it alone unless you're sending a separate vertical feed for Shorts.

Send it to Cameron

Paste the stream key into a text or email to Cameron. Also send the watch link (see *Share the watch link* under Going live and ending).

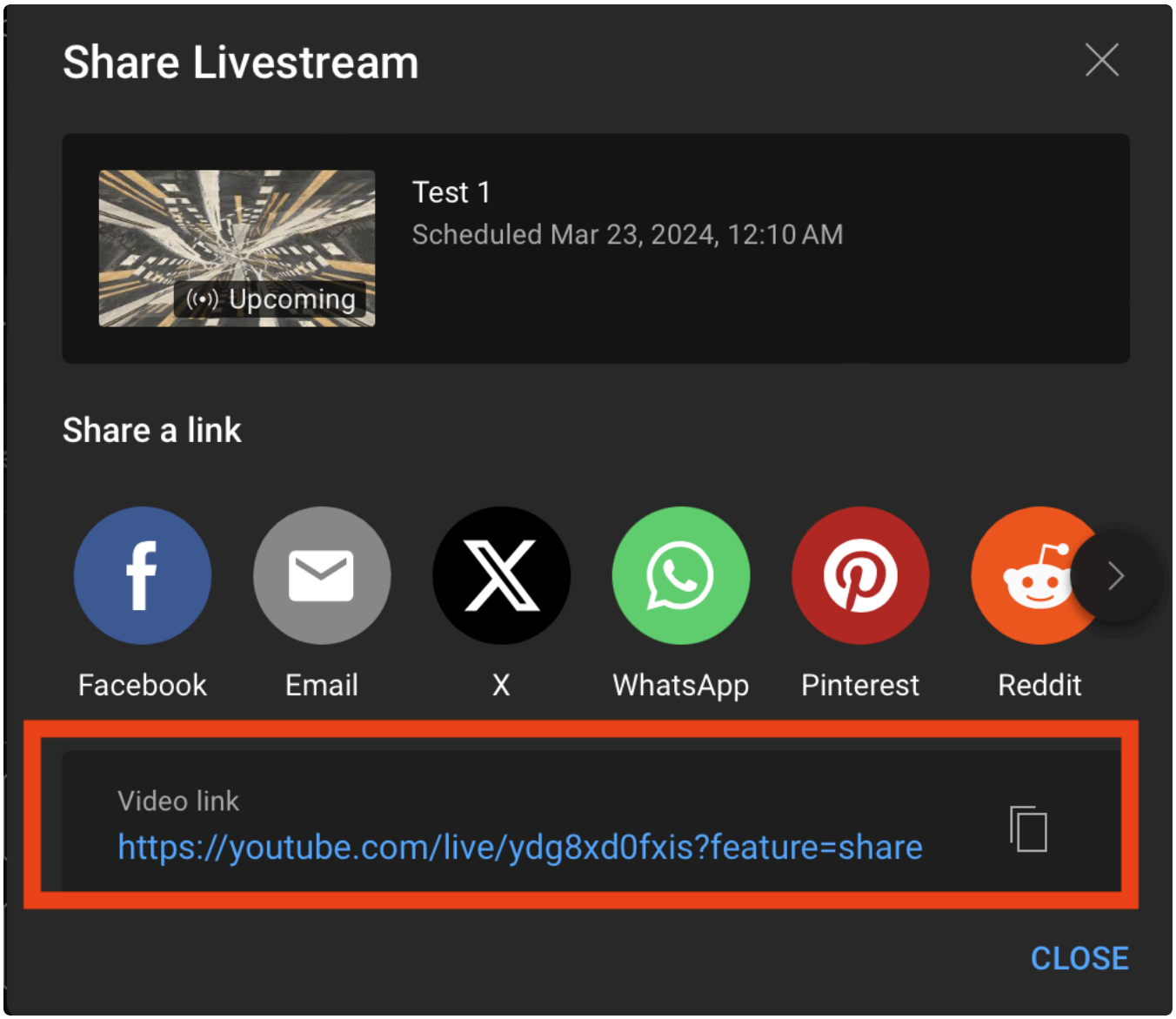
A saved key stays the same from event to event. The default one-time key can change after each broadcast, which is the most common reason a stream that worked last week shows nothing today.

Going live and ending

With auto-start and auto-stop on, the whole show runs from the encoder.

1. **Go live:** press **Start Streaming** (OBS) or **On Air** (ATEM). The broadcast goes live within about 10–30 seconds.
2. **End:** press **Stop Streaming** / **Off Air**. YouTube ends the broadcast and starts processing the replay.
3. **Next event:** in Manage, schedule a new stream with **Reuse settings**. Each broadcast is used once, so every event needs its own scheduled stream, but the key in your encoder stays the same.

Share the watch link. Before the event, click the share arrow (top right of the Control Room) and copy the **Video link**. That's the link viewers use; it works before, during and after the stream.



Pre-show checklist

- [ ] Stream scheduled from Manage, source set to Streaming software
- [ ] Saved stream key selected (not the default one-time key)
- [ ] Auto-start and auto-stop on
- [ ] Key and URL pasted in the encoder and saved
- [ ] Visibility and audience setting correct
- [ ] Preview checked before starting the encoder

Troubleshooting and key security

Problem	Fix
Encoder says it's streaming but YouTube shows nothing	Schedule the stream from Manage, select your saved key, and confirm auto-start is on
Stream ended on its own mid-event	Auto-stop fired on a dropout; restart via a new scheduled stream, or leave auto-stop off on shaky networks
Worked last week, not today	The default one-time key changed; switch to a saved custom key
Can't find Go live	Live streaming isn't enabled yet, or the 24-hour activation hasn't finished

Keep the key private. Anyone with your stream key can broadcast to your channel. Don't paste it in shared chats or screenshots. If it leaks, create a new key in Stream settings and update your encoder.

Sources

- YouTube Help: Manage live stream settings
- Socialive: streaming to YouTube
- MediaCP: YouTube Live broadcasting changes